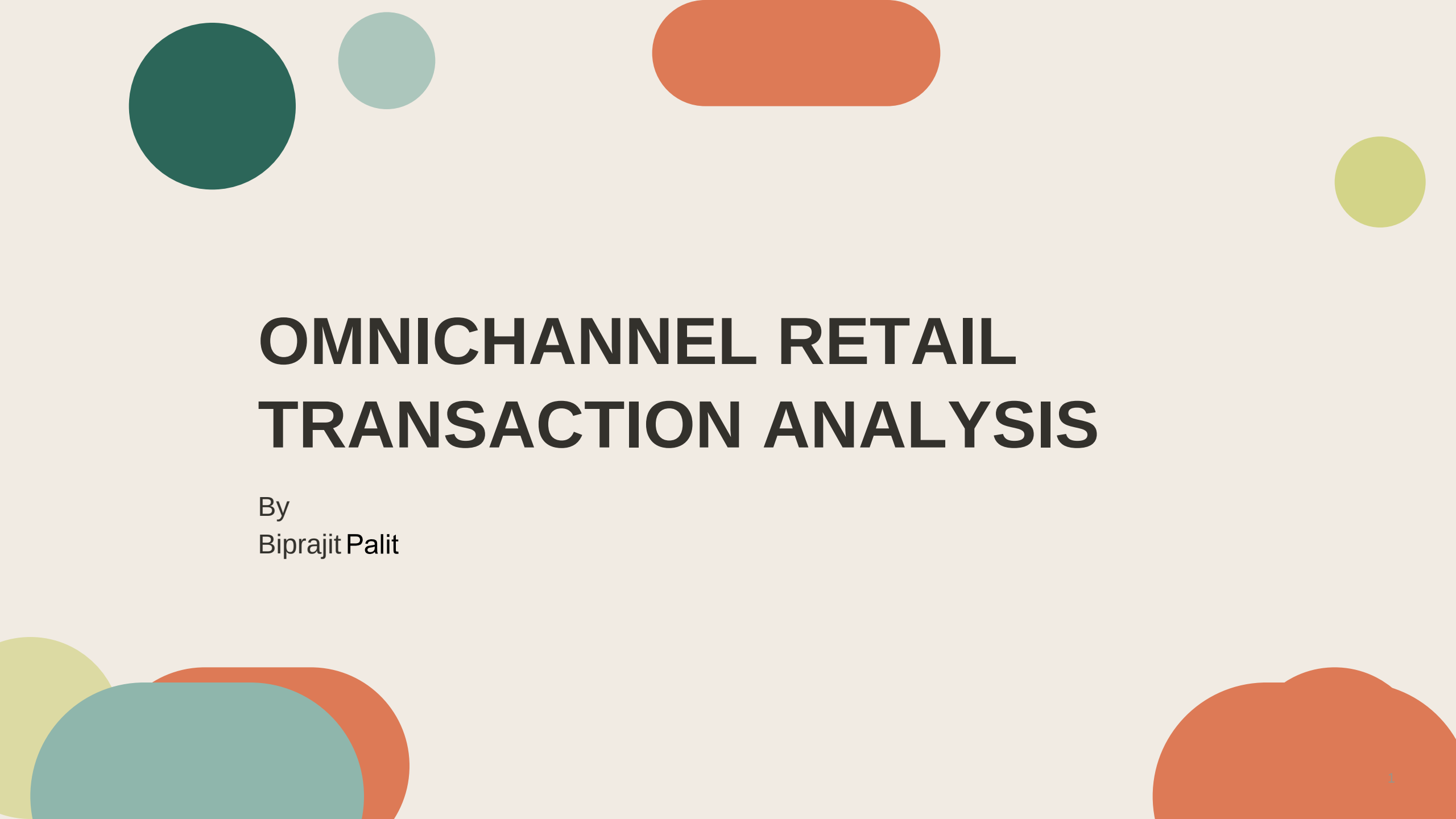




OMNICHANNEL RETAIL TRANSACTION ANALYSIS

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Project Overview

This project showcases essential SQL skills and techniques commonly used by data analysts to explore, clean, and analyze retail transaction data. It involves setting up an omnichannel transaction database, conducting exploratory data analysis (EDA), and using SQL queries to answer key business questions. Built as a hands-on exercise in data analysis, this project helps build a strong foundation in SQL.

Objectives

- **Set up a Transaction Database:** Create and populate a customer_transactions table using the provided retail sales data.
- **Data Sanitation:** Detect and purge records containing missing or null values in critical attributes.
- **Exploratory Data Analysis (EDA):** Conduct baseline EDA to understand record counts, customer footprint, and active product lines.
- **Business Analysis:** Leverage SQL queries to answer key business questions and extract meaningful insights from transaction data.

Project Structure

1. Database Setup

- **Table Creation:** A table named `customer_transactions` is created to store the sales data.
- **Schema:** Columns cover transaction ID, transaction date, transaction time, buyer ID, buyer gender, buyer age, product line, units sold, unit price, cost of goods (COGS), and net revenue.

```
CREATE TABLE customer_transactions
(
    transaction_id INT PRIMARY KEY,
    transaction_date DATE,
    transaction_time TIME,
    buyer_id INT,
    buyer_gender VARCHAR(10),
    buyer_age INT,
    product_line VARCHAR(35),
    units_sold INT,
    unit_price FLOAT,
    cost_of_goods FLOAT,
    net_revenue FLOAT
);
```

Project Structure

2. Data Sanitation & Exploration

- **Record Count:** Determine the total number of records ingested.
- **Customer Count:** Find out how many unique buyers are in the dataset.
- **Category Count:** Identify all unique product lines in the dataset.
- **Null Value Check:** Check for null values across critical attributes and purge incomplete records.

```
-- Sanitation & Exploration
SELECT * FROM customer_transactions
WHERE
    transaction_date IS NULL
    OR buyer_id IS NULL
    OR units_sold IS NULL
    OR unit_price IS NULL ...;

SELECT COUNT(*) AS total_records_ingested
FROM customer_transactions;

SELECT COUNT(DISTINCT buyer_id)
AS distinct_customer_base
FROM customer_transactions;
```

Project Structure

2. Cleaning Results

- **total_records_ingested:** 2,000 raw transaction records loaded.
- **Records purged:** 13 rows removed for missing values in critical fields (mostly buyer_age, units_sold, unit_price, cost_of_goods).
- **total_completed_transactions after cleaning:** 1,987.
- **distinct_customer_base:** 155 unique buyers.
- **active_catalog_categories:** Clothing, Beauty, Electronics.

Data Analysis & Findings

Q1. Retrieve all sales made on transaction_date '2022-11-05'



```
SELECT transaction_id, buyer_id, product_line, net_revenue
FROM customer_transactions
WHERE transaction_date = '2022-11-05';
```

transaction_id	buyer_id	product_line	net_revenue
180	117	Clothing	900
240	95	Beauty	300
1256	29	Clothing	1000
1587	140	Beauty	1200
1819	83	Beauty	100
943	90	Clothing	1200

11 matching transactions in total on this date.

Data Analysis & Findings

Q2. Transactions where product_line = 'Clothing' and units_sold >= 4, in Nov 2022



```
SELECT transaction_id, buyer_id, units_sold, net_revenue
FROM customer_transactions
WHERE product_line = 'Clothing' AND units_sold >= 4
      AND TO_CHAR(transaction_date, 'YYYY-MM') = '2022-11';
```

transaction_id	buyer_id	units_sold	net_revenue
1484	22	4	1200
547	3	4	2000
1476	130	4	2000
735	153	4	2000
943	90	4	1200

17 matching transactions in total.

Data Analysis & Findings

Q3. Total net_revenue for each product_line



```
SELECT product_line, SUM(net_revenue) AS gross_category_revenue
FROM customer_transactions
GROUP BY product_line
ORDER BY gross_category_revenue DESC;
```

product_line	gross_category_revenue
Electronics	311,445
Clothing	309,995
Beauty	286,790

Data Analysis & Findings

Q4. Average buyer_age for the 'Beauty' product line



```
SELECT ROUND(AVG(buyer_age), 2) AS mean_customer_age  
FROM customer_transactions  
WHERE product_line = 'Beauty';
```

mean_customer_age
40.42

Data Analysis & Findings

Q5. High-value transactions where net_revenue >= 1000



```
SELECT transaction_id, buyer_id, product_line, net_revenue
FROM customer_transactions
WHERE net_revenue >= 1000
ORDER BY net_revenue DESC;
```

transaction_id	buyer_id	product_line	net_revenue
592	77	Beauty	2000
1420	138	Clothing	2000
420	28	Clothing	2000
1742	18	Electronics	2000
742	37	Electronics	2000

402 matching transactions in total — a substantial premium-purchase segment.

Data Analysis & Findings

Q6. Transaction volume by product_line and buyer_gender



```
SELECT product_line, buyer_gender,  
       COUNT(transaction_id) AS transaction_volume  
FROM customer_transactions  
GROUP BY product_line, buyer_gender  
ORDER BY product_line, buyer_gender;
```

product_line	buyer_gender	transaction
Beauty	Female	330
Beauty	Male	281
Clothing	Female	347
Clothing	Male	351
Electronics	Female	335
Electronics	Male	343

Data Analysis & Findings

Q7. Peak revenue month per fiscal_year (dense window ranking)

```
WITH monthly_revenue_aggregated AS (  
    SELECT EXTRACT(YEAR FROM transaction_date) AS fiscal_year,  
           EXTRACT(MONTH FROM transaction_date) AS calendar_month,  
           ROUND(AVG(net_revenue), 2) AS average_ticket_size,  
           DENSE_RANK() OVER (...) AS performance_tier  
    FROM customer_transactions GROUP BY 1, 2  
)  
SELECT fiscal_year, calendar_month, average_ticket_size  
FROM monthly_revenue_aggregated WHERE performance_tier = 1;
```

fiscal_year	calendar_month	average_ticket_size
2022	7	541.34
2023	2	535.53

Data Analysis & Findings

Q8. Top 5 buyers by lifetime net_revenue



```
SELECT buyer_id, SUM(net_revenue) AS aggregate_spend
FROM customer_transactions
GROUP BY buyer_id
ORDER BY aggregate_spend DESC
LIMIT 5;
```

buyer_id	aggregate_spend
3	38,440
1	30,750
5	30,405
2	25,295
4	23,580

Data Analysis & Findings

Q9. Unique purchasers reached by each product_line



```
SELECT product_line, COUNT(DISTINCT buyer_id) AS unique_purchasers
FROM customer_transactions
GROUP BY product_line
ORDER BY unique_purchasers DESC;
```

product_line	unique_purchasers
Clothing	149
Electronics	144
Beauty	141

Data Analysis & Findings

Q10. Order throughput by operating shift (Morning <12, Afternoon 12-5, Evening >5)



```
WITH shift_classified_orders AS (  
  SELECT transaction_id,  
  CASE  
    WHEN EXTRACT(HOUR FROM transaction_time) < 12 THEN 'Morning Peak'  
    WHEN EXTRACT(HOUR FROM transaction_time) BETWEEN 12 AND 17  
      THEN 'Afternoon Rush' ELSE 'Evening Drive' END AS operating_window  
  FROM customer_transactions  
)  
SELECT operating_window, COUNT(transaction_id) AS order_throughput  
FROM shift_classified_orders GROUP BY operating_window;
```

operating_window	order_throughput
Evening Drive (>5 PM)	1062
Morning Peak (<12 PM)	548
Afternoon Rush (12-5 PM)	377

Insights

- **Customer Demographics:** Buyers span a broad age range, with transaction volume spread fairly evenly across Clothing, Beauty, and Electronics.
- **High-Value Transactions:** 402 transactions cleared a net_revenue of 1,000 or more, pointing to a meaningful premium-purchase segment.
- **Category Performance:** Electronics leads gross revenue (311,445), narrowly ahead of Clothing (309,995) and Beauty (286,790), despite Clothing reaching the most unique buyers.
- **Sales Trends:** July 2022 and February 2023 stand out as the peak months by average ticket size across their respective years.
- **Shift Behavior:** The Evening Drive window (after 5 PM) accounts for the large majority of order throughput — over half of all transactions.

Conclusion

This project effectively demonstrates how SQL can be used for data analysis in an omnichannel retail context. By setting up a structured `customer_transactions` table, sanitizing and exploring the data, and deriving meaningful business insights, the project covers key aspects of SQL for data-driven decision-making. The findings offer valuable insight into buyer demographics, high-value transactions, sales trends, and purchasing behavior.

The results from this analysis can help a business optimize sales strategies, sharpen customer targeting, and identify peak sales periods and shifts. This project serves as a strong foundation for building SQL skills applicable to data analysis and business intelligence work.



THANK YOU